

SL NO	Monoswini Dey Navonil Mandal DECISION LAYER	DATE 08/10/2025	GOAL	Output
1	Perception Layer (Input → Object Recognition)	Object detection (vehicles, pedestrians, traffic lights, lanes). Object classification (car, bike, bus, human, etc.). Object tracking (track same object across frames → velocity, direction).	Identify and track relevant objects in the environment	Structured data
2	Localization & Lane Estimation Layer	Lane detection (using segmentation or Hough lines). Lane departure detection (check if car crosses lane boundary). Road edge detection (for unmarked roads).	Understand vehicle's position within the lane or road boundary.	Lateral offset (distance from lane center). Lane curvature (road turning left/right). Lane departure flag (True/False).
3	Distance & Time-to-Collision (TTC) Estimation Layer	Estimate relative distance using bounding box scaling or depth from stereo/monocular methods. Compute TTC (Time-to-Collision) If $TTC < \text{threshold}$ → trigger warning/braking logic.	Determine how close and how fast objects are approaching.	If $TTC < \text{threshold}$ → trigger warning/braking logic.{object: "car", distance: 8m, TTC: 1.2s, alert: "High Risk"}
4	Collision Avoidance Decision Layer	Forward Collision Warning (FCW). Automatic Emergency Braking (AEB). Adaptive Cruise Control (ACC) trigger logic.	Prevent or minimize collision risk.	Brake command / Alert command.
5	Driver Monitoring Layer (optional for mobile ADAS)	Eye-blink / head pose detection. Distraction detection (using front camera).	Detect driver inattention or fatigue.	"Driver Drowsy" alert.
6	Traffic Sign & Light Recognition Layer	Traffic light color recognition (stop, go). Speed limit sign detection → feed limit to cruise control logic. if FCW_alert and $lane_departure_alert$: $priority_action = "Brake \& Center"$ elif $pedestrian_ahead$: $priority_action = "Brake"$	Recognize and act on traffic signs or lights.	Detected sign/light + corresponding driving rule.
7	Decision Fusion Layer		Combine insights from multiple subsystems (vision, radar, etc.) into unified decision.	Final driving decision → Brake, Steer, or Alert.
8	Actuation Layer	Brake actuator (for AEB). Steering servo (for lane keep). Haptic/vibration/visual/audio feedback.	Execute final command (for hardware setups).	Physical or virtual response → brake pulse, vibration, on-screen alert.